The Auditor's Workbook: Executable Numbers at The Leap Journal

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A respectable journal shows its working. On this site, an article's numbers need not arrive as assertions: the code that produced them can sit in the article, run at publication, and leave its output on the page for any reader — or referee — to inspect. This annex demonstrates the machinery. Everything below is computed when the site builds; nothing is pasted in.

1 A plain computation

The simplest block: Python in, numbers out. Suppose the Erebor hoard stood at 618 million gold pieces and dragon custody yields nothing, while the pre-dragon economy grew at a modest 2 percent. The cost of 171 years of zero-velocity custody:

```python
hoard = 618_000_000           # gold pieces, T.A. 2770
years = 171                   # duration of dragon custody
growth = 0.02                 # counterfactual annual growth

counterfactual = hoard * (1 + growth) ** years
forgone = counterfactual - hoard
print(f"Counterfactual value:  {counterfactual/1e9:10.2f} billion")
print(f"Forgone output:        {forgone/1e9:10.2f} billion gold pieces")
```

```
Counterfactual value:       18.26 billion
Forgone output:             17.65 billion gold pieces
```

Nearly nineteen billion in forgone output. The dragon, it must be said, slept well.

2 A computed table

Data frames render as tables automatically. The regional ledger of the reflation, by recipient:

```python
import pandas as pd

ledger = pd.DataFrame({
    "Recipient": ["Kingdom under the Mountain", "Dale reconstruction",
                  "Lake-town relief", "Fourteenth share (audited)",
                  "Eagles' consideration"],
    "Share (%)": [52.0, 22.0, 14.5, 7.1, 4.4],
    "Disbursed by": ["Dáin II", "Bard", "Master's office", "This author", "In kind"],
})
```

```
ledger["Gold (millions)"] = (ledger["Share (%)"] / 100 * 618).round(1)
ledger
```

	Recipient	Share (%)	Disbursed by	Gold (millions)
0	Kingdom under the Mountain	52.0	Dáin II	321.4
1	Dale reconstruction	22.0	Bard	136.0
2	Lake-town relief	14.5	Master's office	89.6
3	Fourteenth share (audited)	7.1	This author	43.9
4	Eagles' consideration	4.4	In kind	27.2

3 An interactive chart, in house style

The standard journal theme applies itself: jagged lines, dots at every data point, hover crosshairs, the wordmark, and a download button that exports the graph complete with title, legend, axes and citation.

```
import sys; sys.path.insert(0, "../../theme")
import leaptheme
import plotly.express as px

years = list(range(2935, 2952))
velocity = [0.31, 0.30, 0.31, 0.29, 0.30, 0.28, 0.62,
            1.85, 2.41, 2.20, 1.96, 1.78, 1.66, 1.58, 1.52, 1.49, 1.47]
fig = px.line(x=years, y=velocity,
              labels={"x": "Third Age year", "y": "Velocity of gold (turnover/yr)"})
leaptheme.watermark(fig, article="The Auditor's Workbook",
                    citation="Baggins (2026)",
                    title="The reflation of 2941: velocity of gold")
leaptheme.show(fig, filename="velocity-of-gold-2941")
```

```
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```

(a) Monetary velocity around the reflation of 2941. Fictional data, as is traditional in this journal.

```
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```

(b)

Figure 1

4 A static plot

Not every figure needs interactivity. Matplotlib output lands as a plain image — which also travels into the article's PDF:

```
import matplotlib.pyplot as plt

pops = ["Elves", "Men", "Dwarves", "Hobbits"]
meals = [2, 3, 4, 6.5]
contentment = [6.1, 4.8, 5.5, 9.2]

fig_m, ax = plt.subplots(figsize=(7, 4))
```

```
ax.scatter(meals, contentment, s=90, color="#2e5d43", zorder=3)
for p, x, y in zip(pops, meals, contentment):
    ax.annotate(p, (x, y), textcoords="offset points", xytext=(8, 4))
ax.set_xlabel("Meals per day")
ax.set_ylabel("Contentment (0-10)")
ax.grid(color="#e4e0d5", zorder=0)
for side in ("top", "right"):
    ax.spines[side].set_visible(False)
plt.tight_layout()
plt.show()
```

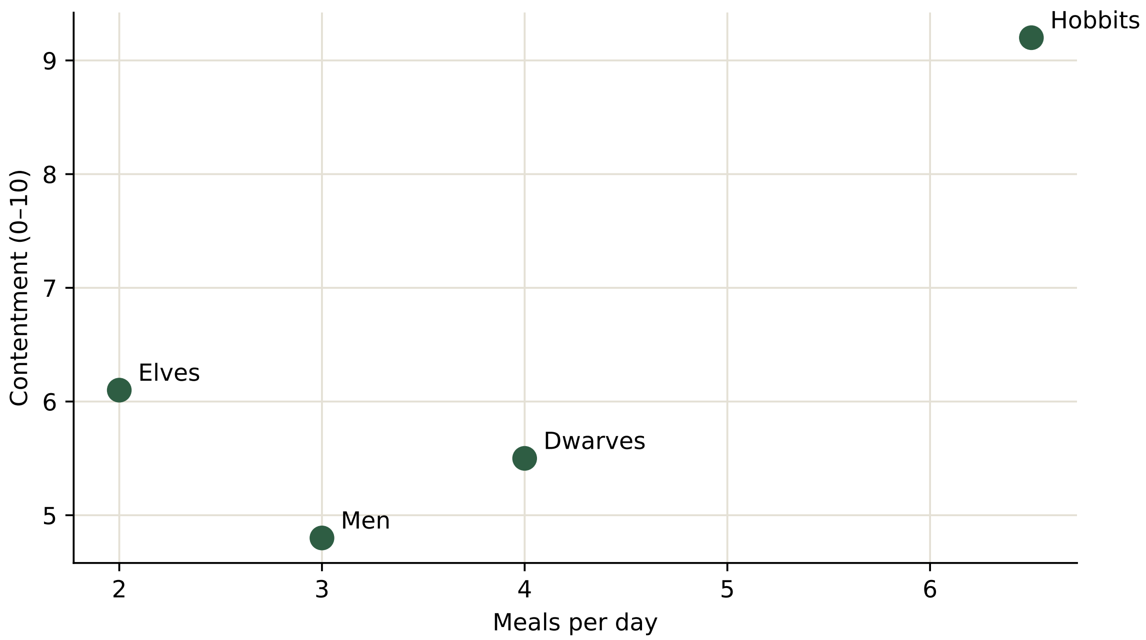

Figure 2: Meals per day and observed contentment across selected populations. Method: asking.

5 A model the reader can adjust

Observable JS blocks run in the reader's browser, so the reader can hold the slider. Set the dragon's counterfactual growth rate yourself and watch the forgone-output estimate move:

(The adjustable model is available in the web version of this article.)

6 The point

None of this required special tooling from the author beyond writing the code in the article file. When the numbers are wrong, the code is wrong, visibly — and a correction is a one-line edit that republishes the article, its charts, its tables, and its PDF together. Auditing, at last, made pleasant.